

Practical Lab File for

Agentic AI

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IMPACT OF ARTIFICIAL INTELLIGENCE IN HEALTHCARE

Introduction:

Artificial Intelligence (AI) is transforming the healthcare industry by enabling faster diagnosis, improving treatment accuracy, and enhancing patient care. AI technologies such as machine learning, natural language processing, and computer vision are being integrated into medical systems to assist healthcare professionals. From predicting diseases to automating administrative tasks, AI is playing a crucial role in making healthcare more efficient, accessible, and cost-effective.

Key Findings:

- AI enables **early disease detection**, especially in conditions like cancer and heart diseases.
- Machine learning models improve **diagnostic accuracy** by analyzing medical images and patient data.
- AI-powered systems assist in **robotic surgeries**, increasing precision and reducing human error.
- Chatbots and virtual assistants enhance **patient engagement and support**.
- AI helps in **drug discovery and development**, reducing time and cost.
- Predictive analytics allows hospitals to **optimize resource management** and patient care.

Challenges:

- **Data Privacy & Security:** Handling sensitive patient data raises serious privacy concerns.
- **High Implementation Cost:** AI systems require significant investment in infrastructure and training.
- **Lack of Skilled Professionals:** There is a shortage of experts who can develop and manage AI systems.
- **Ethical Issues:** Decision-making by AI can raise questions about accountability and trust.
- **Integration Problems:** Difficulty in integrating AI with existing healthcare systems.

Future Scope:

- Development of **AI-based personalized medicine** tailored to individual patients.
- Growth of **smart hospitals** with fully automated systems.
- Wider use of **AI in remote healthcare and telemedicine**.
- Advanced **predictive healthcare systems** to prevent diseases before they occur.
- Integration of AI with wearable devices for **real-time health monitoring**.

Conclusion:

Artificial Intelligence has the potential to revolutionize healthcare by improving efficiency, accuracy, and accessibility. While there are challenges related to cost, ethics, and data security, the benefits of AI far outweigh the limitations. With continuous advancements and proper implementation, AI will play a vital role in shaping the future of healthcare and enhancing the quality of life for patients worldwide.